

Music Genre Relationships Through Hyperbolic Embeddings

Bachelor Project

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Preface

This report presents the bachelor project completed as part of the Data Science programme at the IT University of Copenhagen during Spring 2026. The project was written by Zofia Brodewicz and Mikolaj Andrzejewski under the supervision of Maria Astefanoaei.

All code and related resources are available at: <https://github.com/mvkolaj/Bachelor>

Abstract

This study investigates whether genre distance measured using hyperbolic embeddings provides statistically significant predictive value for music collaboration success on Spotify. Using the Music Genre Dataset containing 43,661 collaborations across nine markets from 2017-2019, we represent 391 music genres in hyperbolic space using the Lorentz model and compute pairwise distances to quantify genre relationships. Through a three-model ablation study comparing Gradient Boosting models with no genre relationship feature, raw ENAO co-occurrence weights, and hyperbolic distance, we find that hyperbolic distance significantly improves prediction accuracy, increasing test R^2 from 0.665 to 0.684 ($p < 0.001$), while raw co-occurrence weights provide no improvement ($p = 0.778$). Despite a near-zero linear correlation with streaming success ($r = -0.03$), genre distance contributes approximately 10% to model performance, ranking third behind market context and source genre popularity. The effect is non-linear and context-dependent: distance reduces prediction error 20 times more for stylistically diverse collaborations than for similar ones, and provides greatest value for Rock collaborations (+24% R^2 improvement) compared to other genre families. These findings demonstrate that hyperbolic embeddings capture meaningful hierarchical genre relationships that enhance collaboration success prediction beyond popularity and market information alone, with predictive value derived from the geometric structure of the learned embeddings rather than the raw co-occurrence patterns used to train them.

1 Introduction

Every year, a handful of unexpected musical collaborations quietly become cultural moments, like a rap artist appearing on a country track, a classical composer co-producing a hip-hop album. These moments are no longer anomalies. They reflect a fundamental shift in how the music industry operates, driven by the rise of digital streaming and increasing competition among artists.

With over 100 million tracks available across major streaming platforms such as Spotify and Apple Music, maintaining visibility and attracting new audiences have become more challenging than ever. Approximately 112,000 new songs are uploaded to these services every day, and 86% of all tracked songs have never surpassed 1,000 streams [2], making discoverability one of the biggest challenges for artists today.

In response, artists have adopted new creative and strategic approaches, with musical collaboration emerging as one of the most prominent. Whether through features, co-writing, or production, collaborations today serve not only as a creative choice but as a tool for expanding reach and increasing commercial success. Cross-genre collaborations in particular have become increasingly common, allowing artists to engage with broader and more diverse audiences than they might reach independently.

These collaborations vary not only in popularity but also in the degree of similarity between the genres involved, referred to here as genre distance. While prior research and industry evidence suggest that streaming success is shaped by factors such as playlist exposure, platform promotion, and existing artist visibility [3], the role of genre-related characteristics has received comparatively little attention. It remains an open question whether genre distance provides additional predictive value for collaboration success beyond other known factors.

This study addresses this gap by examining the following research question: Does genre distance derived from genre hyperbolic embeddings provide statistically significant predictive value in collaboration success prediction? Collaboration success is defined as the average number of streams achieved by collaborations involving artists from given genre pairs. The aim is to evaluate whether including genre distance in predictive models improves their performance, thereby contributing to a deeper understanding of how genre differences influence success in the modern music landscape. These insights could help artists and producers make more informed collaboration decisions, and support streaming platforms in refining how they recommend music to listeners.

2 Related Work

This work is situated within the field of Music Information Retrieval (MIR), which focuses on extracting meaningful patterns from music data, including artist collaborations and genre relationships. Within this domain, understanding the factors that drive musical success has been a central research topic.

A large amount of research shows that collaborations significantly influence song popularity. Ordanini et al. [4] introduce the featuring phenomenon, showing that collaborations between artists from different genres can increase a song’s popularity. Their approach, however, treats genre differ-

ence as a categorical distinction between predefined genre labels rather than modeling genres within a continuous relational space. As a result, the study identifies whether collaborators belong to different genres, but does not quantify the degree of similarity or distance between them. McKenzie et al. [5] similarly find that collaborations between artists from different genre categories are associated with higher streaming demand. However, their analysis also uses broader categorical distinctions to model genre relationships, primarily focusing on whether artists belong to the same or different genres. Additionally, the study considers only a small number of broad genre categories, limiting the level of detail with which genre relationships can be analyzed.

Network-based approaches further support this view, with Silva et al. [6] demonstrating that more connected and central artists tend to achieve greater success. Extending this perspective to the genre level, Oliveira et al. [7] construct genre-based collaboration networks from Spotify data and demonstrate that cross-genre interactions play an increasingly important role in producing successful songs. However, these approaches typically treat genres as categorical entities and do not explicitly model relationships between them.

Although genres are often represented as discrete labels, prior research suggests they form structured systems. Pachet and Cazaly [8] show that genre taxonomies are inherently hierarchical, with broad categories branching into more specific subgenres. This structure is further illustrated by Every Noise at Once [9], an algorithmically-generated map of over 6,000 Spotify genres constructed from listening patterns, artist similarity data and Spotify audio features (like tempo and energy).

Empirical studies further show that certain genre combinations are more likely to be associated with successful songs [10], reinforcing the importance of genre relationships in collaboration outcomes. However, most existing approaches rely on co-occurrence statistics or categorical representations, limiting their ability to capture continuous and hierarchical relationships between genres.

To better represent such structures, recent work has explored embeddings in hyperbolic space. Nickel and Kiela [11] demonstrate that hyperbolic embeddings capture hierarchical relationships more effectively than Euclidean methods. Extending this approach, Nickel and Kiela [12] show that the Lorentz model provides improved numerical stability and more efficient optimization during embedding training compared to the Poincaré-ball formulation, particularly by reducing instability caused by embeddings near the boundary of hyperbolic space. From a theoretical standpoint, Ganea et al. [13] further demonstrate the suitability of hyperbolic space for representing hierarchical structures. In the music domain, Astefanoaei and Collignon [14] apply hyperbolic embeddings to music genre taxonomy by modeling genre co-occurrence data in hyperbolic space. Their results show that hyperbolic embeddings outperform Euclidean embeddings in genre classification tasks by more effectively capturing hierarchical relationships among them. In a broader recommendation setting, Schmeier and Garrett [15] apply hyperbolic embeddings to heterogeneous music graphs containing entities such as artists and tracks, demonstrating that hyperbolic representations better preserve structural relationships for recommendation tasks compared to Euclidean embeddings. While these studies validate hyperbolic embeddings as a representation tool, they primarily focus on tasks such as classification and recommendation rather than using embedding-derived measures in predictive modeling.

In parallel, predicting music success has been widely studied using statistical and machine learn-

ing approaches. Askin and Mauskapf [16] show that both musical features and network position contribute to chart success, while more recent work applies machine learning models to improve predictive performance. Saragih [17] finds that ensemble methods such as Random Forest perform well when using Spotify audio features (e.g., danceability, energy, and tempo) to predict song popularity. Similarly, Lima et al. [18] use gradient boosting models to identify key predictors of musician success by combining features from collaboration networks (such as degree centrality and connectivity measures) with genre-based network features. Their results show that collaboration-network centrality is one of the strongest predictors, highlighting the importance of an artist’s position in music networks. In this context, XGBoost [19] is widely used for tabular predictive tasks due to its ability to model complex nonlinear relationships. However, existing models typically treat genre as a categorical or aggregated feature and do not incorporate continuous representations of genre relationships.

In summary, prior work establishes that collaborations influence success, that genre relationships play a role in these interactions, and that hyperbolic embeddings provide an effective way to model hierarchical genre structures. However, existing approaches do not integrate continuous genre relationships derived from hyperbolic embeddings into predictive models of collaboration success. In particular, the potential of genre distance computed in hyperbolic space as a predictive feature remains largely unexplored. This study addresses that gap by representing genres in the Lorentz model of hyperbolic space and incorporating the resulting genre distance as a predictive feature, evaluated through an ablation study and statistical testing.

3 Data

This section describes the two primary data sources used in this study: the Music Genre Dataset (MGD), which provides collaboration networks and chart performance data, and the Every Noise at Once (ENAO) dataset, which provides genre similarity information. In the following sections we explain how these sources were integrated to construct the final prediction dataset.

3.1 Data Sources

3.1.1 Music Genre Dataset: MGD

Music Genre Dataset (MGD) [33] provides comprehensive data on music collaborations and streaming performance at the genre level. The dataset was constructed by identifying artist collaborations on Spotify, mapping each artist to their primary genre, and aggregating collaboration patterns and chart performance at the genre-pair level.

Coverage:

- **Temporal:** 2017–2019, covering 3 years.
- **Geographic:** 9 markets:
 - Global, worldwide aggregation.
 - Regional: United States (US), United Kingdom (GB), Japan (JP), Australia (AU), Brazil (BR), Canada (CA), Germany (DE), and France (FR).

- **Scale:** Network sizes vary by year:
 - 2017: 258 genres,(or nodes), and 2,643 collaborations, (or edges).
 - 2018: 295 genres and 2,856 collaborations.
 - 2019: 310 genres and 2,928 collaborations.
- **Total edges:** 8,427 unique genre-pair collaborations across all years.

The dataset contains two main components: collaboration networks and chart performance data.

Collaboration Networks: Each row in the collaboration network represents one collaboration between two genres in a specific market and year.

Field	Type	Description
source	string	First collaborating genre, for example “pop”.
target	string	Second collaborating genre, for example “rock”.
market	string	Geographic market, for example “us” or “global”.
year	integer	Year of collaboration: 2017, 2018, or 2019.
weight	float	Number of individual artist collaborations aggregated.
avg_streams	float	Average streaming count for songs in this collaboration.

Table 1: Structure of the MGD collaboration network data.

The dataset structure reflects several important characteristics of how collaboration data is organized. Collaborations are recorded as directed pairs, source $\rightarrow$ target, though the relationship is inherently symmetric: a collaboration between pop and rock is the same whether recorded as pop $\rightarrow$ rock or rock $\rightarrow$ pop. The weight field indicates how many artist collaborations are represented by this genre pair. The avg_streams value represents mean streaming performance across all songs in this collaboration category, providing a robust measure of success that accounts for multiple tracks.

An important feature of the dataset is that market coverage overlaps. A single collaboration can appear in multiple geographic markets simultaneously; for example, a pop $\times$ rock collaboration might appear in both the “global” market and the “us” market with different streaming counts, reflecting how the same musical partnership performs differently across regions. This structure lets us compare market-specific patterns for the same collaboration.

Chart Performance Data:

The MGD dataset also includes Spotify’s Weekly Top 200 Charts for all markets from 2017–2019. These charts rank the 200 most-streamed tracks each week in each market, providing a measure of genre prominence over time.

Coverage:

- **Temporal:** Weekly charts from 2017–2019, approximately 156 weeks.

- **Geographic:** Same nine markets as the collaboration networks.
- **Tracks:** Data for 3,584 unique artists mapped to their genres.
- **Aggregation:** Genre-level chart performance across all weeks.

Chart data structure:

Each weekly chart contains:

- Track rankings from 1 to 200.
- Artist names.
- Streaming counts.
- Market and week identifiers.

Artists were mapped to genres using Spotify metadata included in the MGD dataset. Of the artists in the charts, 3,584 were successfully mapped to genres, while 131 were skipped due to missing genre information.

3.1.2 Genre Similarity Data: ENAO

Every Noise at Once (ENAO) is an algorithmically generated map of music genres and their similarity relationships, created by Glenn McDonald of Spotify [9]. It provides similarity relationships between thousands of music genres based on listening patterns and audio characteristics from Spotify’s catalog.

Dataset structure:

The co-occurrence dataset built from ENAO contains genre similarity information where each genre has a list of similar genres with associated weights ranging from 100 to 240, indicating co-occurrence strength. Relationships are directional but generally symmetric, meaning that if genre A is similar to genre B, the reverse relationship typically exists as well, though potentially with different weights.

Coverage:

- Total genres: 6,283 unique genres in the raw dataset.
- Similarity edges: 123,438 genre-to-genre similarity relationships.

Usage in this study:

We use the ENAO similarity data as the foundation for constructing hyperbolic genre embeddings, described in Section 4. The similarity weights provide the training signal for learning geometric representations that capture hierarchical and semantic relationships between genres. These embeddings enable us to compute meaningful distances between any pair of genres, which serves as our primary measure of genre diversity in collaboration prediction.

3.2 Data Integration and Matching

Combining the two data sources required careful matching of genre names across different naming conventions.

Challenges:

Different datasets used different genre naming conventions:

- **MGD:** Standardized lowercase genre names, such as “pop rap” and “dance pop”.
- **ENAO:** Comprehensive collection of genre labels, including many niche and specially formatted genre names.

Matching process:

We matched genres between MGD collaboration networks and the ENAO embedding vocabulary using exact string matching. For each collaboration in the MGD dataset, we checked whether both the source and target genres existed in the ENAO vocabulary. The final overlap consisted of 391 genres. This represents the working vocabulary for which we have both collaboration data and hyperbolic distance measurements.

3.3 Analysis Dataset

The analysis dataset combines collaboration networks, genre distances, and popularity metrics into a unified table used for both prediction model development and exploratory analysis. Each row represents one collaboration between two genres in a specific market and year, with features capturing hyperbolic distance between genres, chart-based popularity metrics, collaboration frequency, and streaming success.

3.3.1 Dataset Construction Process

We merged the MGD collaboration networks with ENAO embeddings and chart data through the following steps:

1. **Starting point:** Each collaboration from MGD genre networks (source genre, target genre, market, year, weight, avg_streams).
2. **Genre distance computation:** For each collaboration, we computed hyperbolic distance between source and target genres using the ENAO-trained embeddings (methodology detailed in Section 4.)
3. **Popularity metrics:** We derived ranking-based popularity scores for each genre from Spotify’s Top 200 chart data.
4. **Data filtering:** We retained only collaborations where both genres had embeddings and chart popularity data.

This process yielded a final dataset of 43,661 collaborations across nine markets and three years (2017–2019).

3.3.2 Ranking-Based Popularity Metric

Rather than using raw streaming counts as popularity metrics, we developed a ranking-based scoring system to measure genre prominence in each market and year.

Computation process:

For each genre-market-year combination:

1. **Identify chart appearances:** Find all weeks where tracks from that genre appeared in the Top 200.
2. **Convert positions to scores:** For each appearance, convert chart position to points:
 - Rank 1 (most popular) = 200 points
 - Rank 2 = 199 points
 - Rank 200 (least popular) = 1 point
3. **Aggregate across weeks:** Sum all position scores across the year to get total ranking score.
4. **Apply log transformation:**

$$\text{log_ranking_score} = \log_{10}(\text{total_ranking_score} + 1)$$

Rationale for ranking-based approach:

We chose ranking-based popularity over raw streaming counts for three important reasons:

- **Avoids circularity:** Since our target variable is collaboration streaming success, using raw stream counts as a predictor would create logical circularity. Ranking scores provide an independent measure of genre popularity based on chart position rather than absolute streams.
- **Reduces sparsity:** Many genres appear in the Top 200 charts only occasionally or briefly. Raw streaming counts for these genres would have many zero or missing values across weeks. The ranking-based score aggregates all chart appearances into a single cumulative measure. This provides a meaningful popularity metric even for genres with limited chart presence. For example, a genre that appears at rank 150 for only three weeks still receives a measurable score of 150 points. In contrast, raw stream counts would contain mostly zeros with only a few sporadic observations.
- **Reduces scale sensitivity:** Rankings convert stream counts to ordinal positions, making the metric less dependent on market size than raw streams.

3.3.3 Final Dataset Structure

Dataset size:

- Total: 43,661 collaborations

- Train set: 34,929 rows (80%, stratified by market)
- Test set: 8,732 rows (20%, stratified by market)

Feature descriptions:

Feature	Type	Description	Range	Example
market	categorical	Geographic market	9 values	“us”, “global”
year	integer	Year of collaboration	2017–2019	2018
source	string	First genre in collaboration	391 genres	“pop”
target	string	Second genre in collaboration	391 genres	“rock”
distance	float	Hyperbolic distance (Lorentz)	[0.02, 9.20]	4.35
log_ranking_score_source	float	$\log_{10}(\text{ranking_score_source} + 1)$	[0, ~ 5]	4.39
log_ranking_score_target	float	$\log_{10}(\text{ranking_score_target} + 1)$	[0, ~ 5]	4.28
log_weight	float	$\log_{10}(\text{weight} + 1)$	[0.3, 2.7]	1.08
log_streams	float	$\log_{10}(\text{avg_streams} + 1)$	[5.4, 8.9]	6.7

Table 2: Feature descriptions for the final analysis dataset.

Example rows:

market	year	source	target	distance	log_source	log_target	log_weight	log_streams
au	2017	dance pop	pop	3.045	5.597	5.814	2.400	6.769
au	2017	pop	pop rap	6.245	5.814	5.304	2.340	6.665
au	2017	pop rap	rap	0.151	5.304	5.322	2.283	6.495

Table 3: Example rows from the final dataset. *log_source* and *log_target* refer to log-transformed ranking scores.

3.3.4 Feature Engineering Decisions

Log transformations:

We log-transform all continuous variables (popularity scores, weight, streams) for the following reasons:

- **Skewed distributions:** Music data exhibits heavy right skew, a few mega-hits dominate while most content has modest performance.
- **Multiplicative relationships:** Popularity and success likely interact multiplicatively. For example, doubling both genres’ popularity might more than double collaboration success.
- **Scale normalization:** Log transformation brings variables to comparable scales, improving model stability.
- **Variance stabilization:** Log transformation reduces heteroscedasticity (non-constant variance), making model residuals more homogeneous across the prediction range. Without transformation, prediction errors tend to be larger for high-success collaborations than for low-success ones. Applying the log transformation makes prediction errors more consistent across success levels, improving the validity of statistical tests and confidence intervals.

Market encoding:

Markets are label-encoded as integers (0–8) for tree-based models, which handle categorical variables

naturally without requiring one-hot encoding.

Target variable selection:

We predict *log_streams* rather than raw *avg_streams* because:

- Log scale linearizes multiplicative effects
- Reduces influence of extreme outliers (mega-hits)
- Model errors are more interpretable (error in log space $\approx$ percentage error in original space)

Data quality:

The dataset contains no missing values or duplicates, as incomplete data was filtered during construction and each market-year-source-target combination appears exactly once. To ensure balanced geographic representation, the train/test split was stratified by market so all regions are proportionally represented in both training and test sets.

4 Methods

This section describes the methodology used to compute genre distances and evaluate their predictive value for collaboration success.

4.1 Hyperbolic Genre Embeddings

To capture the hierarchical structure inherent in music genre taxonomies, we employed hyperbolic embeddings. Hyperbolic space is particularly well-suited for representing hierarchical data because it can embed tree structures with minimal distortion, unlike Euclidean space where the number of nodes grows exponentially with depth [13],[22].

4.1.1 Adjacency Matrix Construction

We begin by constructing a weighted adjacency matrix $\tilde{A}$ from the ENAO genre similarity data, which serves as the input for training hyperbolic embeddings. The raw ENAO dataset provides similarity scores between genre pairs, which we organize into a directed graph representation.

Process:

From the ENAO dataset containing 6,283 genres, we extract 123,438 weighted edges representing similarity relationships, where each edge weight indicates the strength of similarity between two genres. We organize these edges into a square adjacency matrix and remove self-loops (i.e., collaborations between the same genres). To make the weights comparable, we normalize each row so that all outgoing connections from a genre sum to 1. We verify that all genres form a single connected component, though we remove three isolated genres with no outgoing connections (“lgbtq+ hip hop”, “pop lgbtq+ brasileira”, and “re:techno”). This yields a final normalized adjacency matrix of 6,280 genres by 6,280 genres, ready for embedding training.

4.1.2 Lorentz Model for Hyperbolic Embeddings

We use the Lorentz model of hyperbolic space to represent genre embeddings. In this model, each genre is represented by coordinates $\mathbf{u}_i = (t_i, x_i, y_i, z_i)$ in four-dimensional space, constrained to lie on a hyperbolic surface. The distance between two genres is computed using the Lorentz distance formula:

$$d_L(\mathbf{u}_i, \mathbf{u}_j) = \text{arcosh}(t_i t_j - x_i x_j - y_i y_j - z_i z_j) \quad (1)$$

This distance naturally captures both hierarchical relationships and semantic similarity between genres, Nickel and Kiela [12].

Embedding training:

We train the embeddings by optimizing them to preserve the similarity structure encoded in the adjacency matrix $\tilde{A}$. The training process uses Riemannian optimization on the Lorentz manifold [23], which ensures that embeddings remain on the hyperboloid throughout training.

The model learns to place similar genres close together in hyperbolic space. During training, the model minimizes the distance between genre pairs that have high similarity in the ENAO data, while keeping dissimilar genres far apart. This is achieved through negative sampling, a technique that compares real genre similarities against random non-similar pairs.

Implementation:

We use the Lorentz genre embeddings implementation [21], which builds on the framework of Nickel and Kiela [12]. The training hyperparameters are:

- Embedding dimension: $d = 3$, resulting in 4D Minkowski coordinates.
- Learning rate: 0.1.
- Negative samples: 10 per positive edge.
- Epochs: 500 with early stopping based on validation loss.
- Optimizer: Riemannian Adam [23].

4.1.3 Distance Computation for Collaboration Dataset

For each collaboration in our dataset, we compute the Lorentz distance between the source and target genres using their learned hyperbolic embeddings.

Genre 1	Genre 2	Distance	Interpretation
pop	dance pop	3.05	Close, subgenre relationship
pop	rock	4.82	Moderate, different but related
pop	death metal	8.51	Far, stylistically distant
rap	pop rap	0.15	Very close, direct subgenre
hip hop	rap	1.23	Close, overlapping genres

Table 4: Example hyperbolic distances between selected genre pairs.

These distances capture both hierarchical relationships and semantic similarities. As shown in Table 4, the small distance between “rap” and “pop rap”, 0.15, reflects that pop rap is a direct combination of rap with pop elements, while the large distance between “pop” and “death metal”, 8.51, reflects their extreme stylistic differences.

Across all 43,661 collaborations in our dataset, hyperbolic distances range from 0.02 to 9.20, with a mean of 4.19 and standard deviation of 1.74. This variation in distance provides the foundation for testing whether genre similarity influences collaboration success.

4.1.4 Embedding Visualization and Validation

To verify that the trained embeddings capture meaningful genre relationships, we visualize them using the Poincaré disk model and analyze their geometric properties.

Poincaré disk projection:

While we train our embeddings using the computationally efficient Lorentz model, we can map them to the Poincaré disk for intuitive visualization [11]. The Poincaré disk represents hyperbolic space as a unit disk, where the center corresponds to more general genres, the boundary to more specific ones, and distance from the origin encodes hierarchical level.

Full ENAO Embedding Space:

Figure 1 shows all 6,280 genres mapped to the Poincaré disk. The visualization reveals several key properties of the embedding space. Most genres cluster near the boundary of the disk, indicating they are specific subgenres rather than broad categories, while the center is relatively sparse, suggesting few genres occupy top-level parent positions in the hierarchy. Genres are not uniformly distributed but form clusters at various radii, reflecting different levels of the genre taxonomy.

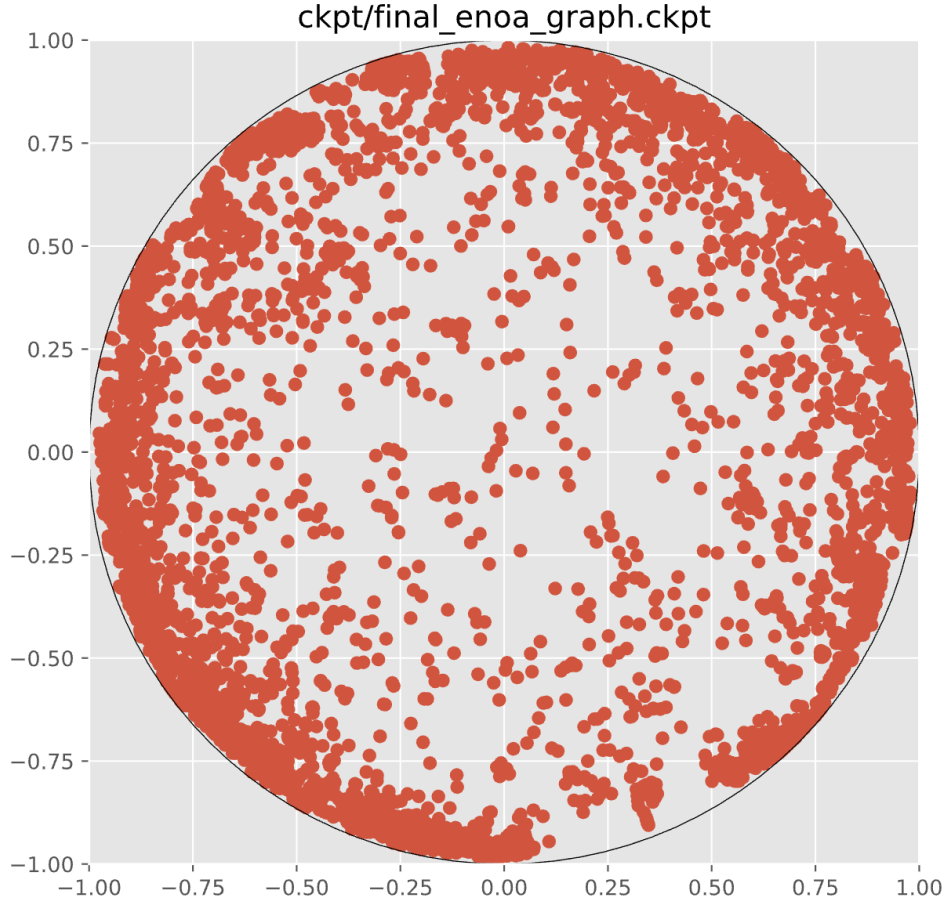


Figure 1: Full ENAO embedding space visualized in the Poincaré disk. All 6,280 genres are projected from Lorentz coordinates into the disk representation.

Genre Family Clusters

To better understand the semantic structure, Figure 2 highlights selected genres from five major families we identified from the ENAO website: Pop (blue), Rap (orange), Rock (green), Urbano Latino (purple), and Hip Hop (red). These families represent the most popular genre groupings and are based on stylistic similarity and cultural affinity as observed in Spotify listening patterns.

The visualization reveals several key properties of how genres organize in hyperbolic space:

- **Family cohesion:** Genres within the same family cluster spatially. For example, rock subgenres, including indie rock, alternative rock, classic rock, and folk rock, occupy the upper-right quadrant.
- **Hierarchical positioning:** More general genres are closer to the center, while specific subgenres are pushed toward the periphery.
- **Cross-family relationships:** Some genre families are positioned close to each other, indicating stylistic overlap. For instance, Rap and Hip Hop families are adjacent in the bottom half of the disk, reflecting their close musical relationship.

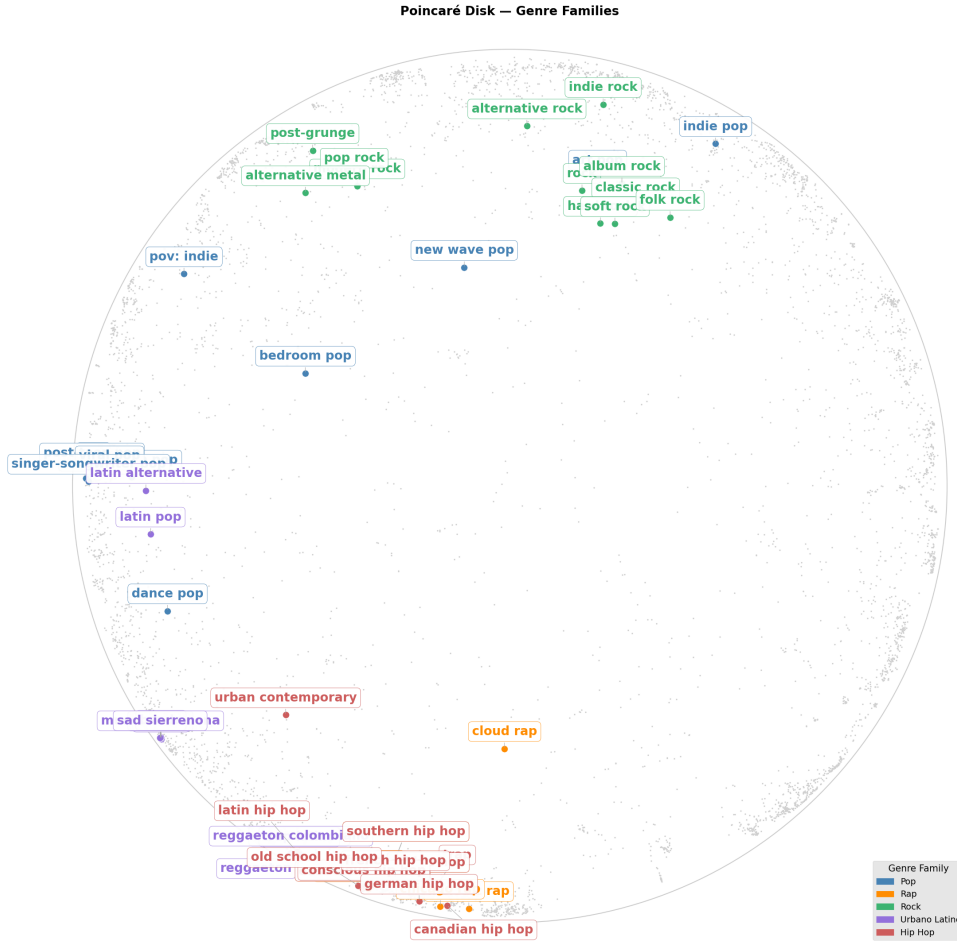


Figure 2: Poincaré disk visualization of selected genre families. Pop, Rap, Rock, Urbano Latino, and Hip Hop genres form semantically meaningful clusters in hyperbolic space.

The visualization also reveals several unexpected patterns that highlight limitations in the ENAO co-occurrence dataset. For example, cloud rap appears closer to the center than other rap subgenres, suggesting a more general hierarchical position despite its stylistic specificity. Similarly, some niche subgenres appear at unexpected radii or positions relative to their parent genres. These anomalies likely reflect biases in the underlying co-occurrence data. Genres with limited playlist representation or inconsistent tagging may produce embeddings that do not fully capture their true position in the genre taxonomy.

Intra-Family Distance Analysis

To quantify the cohesiveness of genre families in the embedding space, Figure 3 compares the average distance between genres within each family (intra-cluster distance) against the global average distance across all genre pairs (5.734). All five families show intra-cluster distances well below this global average, indicating that the embeddings successfully capture genre family structure. The fact that genres within a family are significantly closer than random genre pairs validates that hyperbolic distance is a meaningful measure of genre similarity.

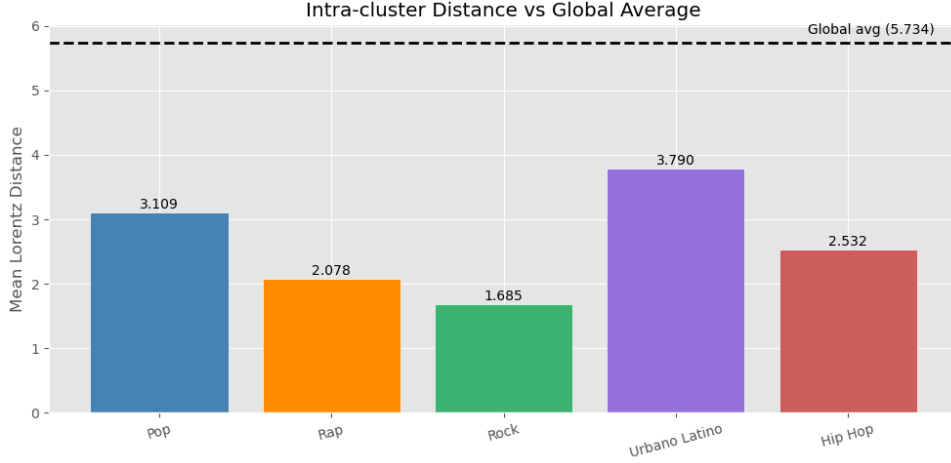


Figure 3: Intra-family distance compared with the global average distance. All selected genre families show substantially lower internal distances than the global average, indicating that the embeddings capture meaningful genre-family structure.

4.1.5 Baseline: Direct ENAO Co-occurrence Weights

To provide a comparison point for our hyperbolic embeddings, we extract raw ENAO edge weights as a baseline feature. For each genre pair (a, b) , we compute their similarity as the average of bidirectional edge weights from the normalized adjacency matrix:

$$\text{similarity}(a, b) = \frac{A[a, b] + A[b, a]}{2} \quad (2)$$

where A is the row-normalized ENAO adjacency matrix. Each edge weight reflects how strongly two genres are connected, based on listening patterns and audio characteristics. Taking the average of both directions ensures symmetry, so that the similarity between genre a and genre b equals the similarity between genre b and genre a .

This provides a direct measure of genre co-occurrence strength, serving as a natural baseline. It allows us to test whether the predictive value of hyperbolic embeddings comes from the geometric structure they learn, or simply from the co-occurrence patterns in the data they were trained on. The key difference is that while hyperbolic distance captures the full hierarchical structure of the genre taxonomy, ENAO similarity only reflects whether two genres are directly connected, with no information about their broader relationships to other genres.

4.2 Feature Engineering

Our prediction models use six features derived from the analysis dataset described in Section 3.3.

Feature	Type	Description	Range
distance	float	Hyperbolic distance between genres	[0.02, 8.94]
log_ranking_score_source	float	$\log_{10}(\text{ranking score} + 1)$ for source genre	[0, ~ 5]
log_ranking_score_target	float	$\log_{10}(\text{ranking score} + 1)$ for target genre	[0, ~ 5]
log_weight	float	$\log_{10}(\text{collaboration frequency} + 1)$	[0.3, 2.7]
market_enc	int	Market identifier (label encoded)	$\{0, 1, \dots, 8\}$
year	int	Year of collaboration	$\{2017, 2018, 2019\}$

Table 5: Features used for prediction models.

The target variable is log-transformed average streams:

$$\mathbf{log_streams} = \log_{10}(\text{average streams} + 1)$$

We log-transform all continuous features to handle the heavy right skew in music streaming data, as explained in Section 3.3. Markets are label-encoded as integers (0–8) for tree-based models, which handle categorical variables naturally without one-hot encoding [24],[27]. For Linear Regression, we use one-hot encoded market dummy variables to avoid imposing artificial ordinal relationships.

The dataset is split 80/20 into training (34,928 rows) and test (8,733 rows) sets, stratified by market to ensure balanced geographic representation [26].

4.3 Prediction Models

We evaluate three regression models representing a spectrum from linear to complex non-linear relationships: Linear Regression, Random Forest, and Gradient Boosting. This diversity allows us to test whether genre distance provides value across different model families.

4.3.1 Model Selection

Linear Regression:

Linear Regression serves as our baseline model, assuming log-transformed streams can be expressed as a linear combination of features [27]:

$$\text{log_streams} = \beta_0 + \beta_1 \cdot \text{distance} + \beta_2 \cdot \text{log_pop_source} + \beta_3 \cdot \text{log_pop_target} + \beta_4 \cdot \text{log_weight} + \sum_i \gamma_i \cdot \text{market}_i + \beta_5 \cdot \text{year} + \epsilon$$

Random Forest:

Random Forest is an ensemble of decision trees that can capture non-linear relationships and feature interactions without explicit specification [24]. The final prediction is the average across all tree predictions:

$$\hat{y} = \frac{1}{B} \sum_{b=1}^B \text{Tree}_b(x)$$

where B is the number of trees.

Gradient Boosting:

Gradient Boosting builds an ensemble of trees sequentially, where each new tree corrects the errors of the previous ensemble [25]. The model iteratively minimizes a loss function L by adding trees in the direction of the negative gradient:

$$F_m(x) = F_{m-1}(x) + \nu \cdot h_m(x)$$

where F_m is the ensemble after m iterations, h_m is the new tree, and ν is the learning rate.

4.3.2 Hyperparameter Tuning

For Random Forest and Gradient Boosting, we perform hyperparameter optimization using `RandomizedSearchCV` [28], sampling 20 random parameter combinations with three-fold cross-validation and R^2 as the scoring metric.

4.3.3 Model Evaluation

We evaluate models using R^2 (coefficient of determination), which measures the proportion of variance in collaboration success explained by the model [27]. R^2 ranges from zero to one, where higher values indicate better predictive performance. To assess generalization, we perform five-fold cross-validation on the training set [26], reporting the mean R^2 across folds. For final models, we report both training and test R^2 , where a large gap between the two indicates overfitting.

4.4 Statistical Testing: Ablation Study

4.4.1 Design

An ablation study systematically removes or replaces features to measure their individual contributions [29]. To test whether genre distance provides statistically significant predictive value and whether hyperbolic embeddings improve upon raw co-occurrence data, we design a three-model ablation study. We compare the following variants of our best-performing Gradient Boosting model:

- **Model A (Baseline):** Includes popularity, weight, market, and year only, excludes any genre relationship feature.
- **Model B (ENAO Similarity):** Extends the baseline with raw ENAO co-occurrence weights (`enaosimilarity`).
- **Model C (Hyperbolic Distance):** Extends the baseline with hyperbolic distance from the Lorentz embeddings (`distance`).

All three models use identical hyperparameters and training data. The only difference between them is the presence or absence of a genre relationship feature, allowing us to isolate its contribution to predictive performance. Comparing Model C with Model A isolates the contribution of hyperbolic distance, while comparing Model C with Model B tests whether the geometric structure of hyperbolic embeddings provides additional predictive value beyond the raw co-occurrence weights used to train them.

4.4.2 Statistical Significance Testing

We use paired t-tests to determine whether the performance differences between models are statistically significant [30]. The paired design is appropriate because all models are evaluated on the same 8,733 test samples, controlling for sample-specific variance. For each test sample, we compute squared prediction errors for all three models and calculate pairwise differences. The t-test evaluates whether the average difference across all samples is significantly different from zero, using $\alpha = 0.05$ as the significance threshold [31]. Three hypotheses are tested:

- H_0^{AB} : ENAO similarity provides no additional predictive value over the baseline.
- H_0^{AC} : Hyperbolic distance provides no additional predictive value over the baseline.
- H_0^{BC} : Hyperbolic distance provides no additional predictive value over ENAO similarity.

4.4.3 Effect Size Metrics

Beyond statistical significance, we report effect sizes to quantify practical importance [32]:

- ΔR^2 : Change in explained variance between models.
- Mean error reduction: Average decrease in squared prediction error.
- Percentage improvement: Relative error reduction.

These metrics distinguish between statistical significance (is the effect real?) and practical significance (is the effect meaningful?).

5 Exploratory Analysis

This section examines the distribution of genre distances and their relationship with collaboration success, measured as average streams. The analysis is conducted at three levels of granularity: globally, broken down by individual market, and broken down by genre family.

Genre distances are broadly distributed, with a global mean of 4.19 that remains stable across all three years and nine markets examined. Oliveira et al. [7] show that the number of cross-genre collaborations increased over this period. However, the average hyperbolic distance of those collaborations remains stable, suggesting that while artists collaborate more frequently across genres, they are not reaching further across the genre space over time. At the genre family level, intra-family distances vary considerably, from a mean of 0.88 in Rock to 3.54 in Hip Hop, reflecting meaningful differences in how tightly different genre families cluster in hyperbolic space.

Correlation analysis reveals a statistically significant but very weak negative relationship between genre distance and log streams at the global level (Pearson $r = -0.03$, $p < 0.05$; Spearman $r = -0.03$, $p < 0.05$). Although greater distance is associated with slightly lower streaming numbers, the effect size is negligible, indicating that genre distance alone explains very little variation in streaming performance. Most markets exhibit a similar weak negative trend, with France as a notable exception, showing a small positive correlation ($r = 0.06$). Within genre families, correlations are almost universally non-significant, indicating that no consistent relationship between genre distance and streaming performance is detected at this level of aggregation.

Taken together, these results confirm the absence of a universally strong linear relationship between genre distance and success. However, linear correlation may not fully capture the structure of this relationship. When collaborations are grouped into low, medium, and high distance bins, a consistent inverted-U pattern emerges. Medium-distance collaborations achieve the highest average streams globally (78.2M), outperforming both low-distance (69.4M) and high-distance (67.0M) groups. This pattern appears across most markets and years, suggesting that a moderate degree of genre distance may be commercially optimal.

At the genre family level, however, patterns vary considerably. Hip Hop favors lower distances, with streams decreasing as distance increases, while Latin shows the opposite trend, though with limited observations. Pop and Rap display no consistent pattern across years, and Rock collaborations are almost entirely concentrated in the low-distance bin, making meaningful comparison across groups impossible. These differences suggest that the optimal degree of genre distance is context-dependent rather than universal, consistent with findings by Ordanini et al. [4], who show that genre distance matters more in some genres than others.

Overall, genre distance is not a universally strong linear predictor of success, but it does exhibit meaningful non-linear patterns, suggesting that while distance alone has limited explanatory power in a linear framework, it may still carry predictive value. This motivates its inclusion as a feature in the predictive models examined in the following section.

6 Prediction Model Analysis

To answer our research question we built prediction models to estimate collaboration success. This section presents our model comparison strategy, evaluates different modeling approaches, and analyzes whether distance contributes to prediction value.

6.1 Model Selection and Comparison

We compared three models spanning the spectrum from linear to complex non-linear approaches:

1. **Linear Regression** - Tests whether genre distance has a simple additive effect
2. **Random Forest** - Ensemble of decision trees to capture non-linear patterns
3. **Gradient Boosting** - Sequential ensemble that builds trees iteratively to correct errors

This comparison allows us to determine whether the relationship between genre distance and collab-

oration success is linear or requires more sophisticated modeling to capture complex interactions.

6.1.1 Performance Interpretation

Model	5-Fold CV R^2	Test R^2	Best Parameters
Linear Regression	0.154 (± 0.006)	0.149	$\alpha = 0.1$
Random Forest	0.639 (± 0.005)	0.655	n_estimators = 300, max_depth = 20
Gradient Boosting	0.678 (± 0.001)	0.684	n_estimators = 300, max_depth = 7, learning_rate = 0.1

Table 6: Prediction model performance comparison

The results reveal a clear performance hierarchy. Linear Regression fails dramatically, achieving only $R^2 = 0.149$ on the test set explaining hardly 15% of variance in collaboration success.

In contrast, tree-based models succeed. Both Random Forest ($R^2 = 0.655$) and Gradient Boosting ($R^2 = 0.684$) achieve strong performance, explaining approximately 65-68% of variance. This represents almost a fivefold improvement over linear regression, demonstrating that the relationships present in the data are highly non-linear.

Gradient Boosting achieves the best performance with test R^2 of 0.684. The improvement over Random Forest is modest but consistent across cross-validation folds. The close alignment between CV R^2 (0.678) and test R^2 (0.684) confirms good generalization to unseen data, with minimal overfitting.

The superior performance of tree-based models over Linear Regression suggests the relationship between genre distance and collaboration success is non-linear. Linear Regression assumes that distance has a constant effect across all observations. In contrast, tree-based models can capture how distance’s influence varies depending on context - such as genre popularity, market, or the specific genres involved.

For instance, a distance of 4.0 might mean something different for a Pop $\times$ Dance Pop collaboration compared to a Rock $\times$ Experimental collaboration. Tree-based models can learn these nuanced patterns, whereas linear models apply the same coefficient to all distance = 4.0 pairs regardless of context.

This finding motivates our choice of Gradient Boosting for all subsequent analyses and validates the need for sophisticated modeling approaches when studying genre relationships.

6.2 Model Fit Assessment

To validate whether our best model produces accurate predictions, we examine the relationship between predicted and actual streaming performance.

Figure 4 shows predicted versus actual $\log(\text{streams})$ on the test set:

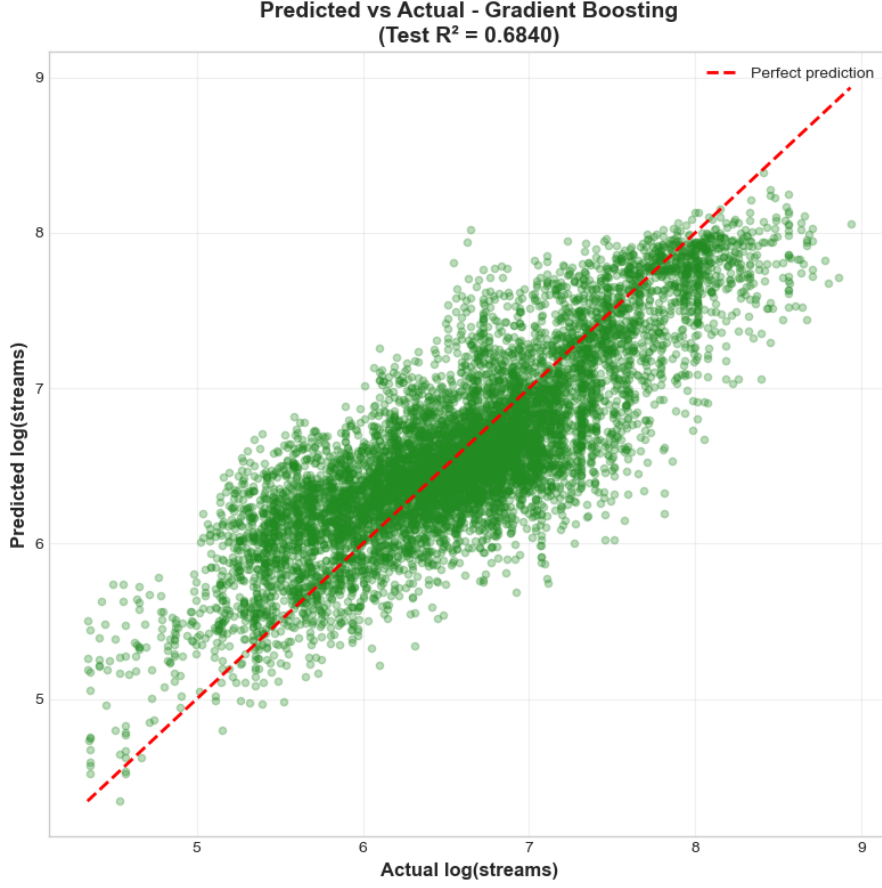


Figure 4: Predicted vs actual collaboration success on the test set. Points cluster along the diagonal, indicating good model fit across the full range of success levels ($R^2 = 0.684$).

The scatter plot demonstrates that the model performs well across the entire spectrum of collaboration success. Points cluster reasonably close to the diagonal reference line, which represents perfect predictions.

Our model explains 68.4% of variance in collaboration success. The remaining 31.6% likely reflects unmeasured factors, including artist-level characteristics (star power, fan base loyalty), song-level features (tempo, key, lyrical content, production quality), promotional efforts (marketing spend, playlist pitching), and inherent randomness (viral moments, social media trends, timing). Despite these limitations, achieving $R^2 = 0.684$ using only genre-level features demonstrates that genre relationships, particularly distance, contain substantial predictive information about collaboration outcomes.

6.3 Feature Importance Analysis

Having established that Gradient Boosting best captures the relationship between features and collaboration success, we now examine which features contribute most to the model’s predictions. Understanding feature importance reveals where genre distance ranks among all the factors we measured. While our research question focuses on whether distance provides significant predictive value, importance scores show its relative contribution compared to popularity, weight, and other features.

In Gradient Boosting, feature importance is computed by summing the total reduction in mean squared error achieved by all splits on that feature across all trees in the ensemble [25]. When building each

tree, the algorithm selects the feature and threshold that maximally reduce prediction error at each node. A feature’s importance reflects both how often it is selected for splits and how much error it reduces when used. The raw importance values are normalized to sum to 1.0, allowing interpretation as the percentage each feature contributes to overall model performance.

6.3.1 Overall Feature Importance

Table 7 presents the detailed breakdown:

Feature	Importance	Interpretation
market_enc	60%	Market-specific patterns dominate
log_ranking_score_source	11%	Source genre popularity
distance	10%	Genre similarity/diversity
log_ranking_score_target	10%	Target genre popularity
log_weight	6%	Collaboration frequency
year	3%	Temporal trends (minimal)

Table 7: Feature importance breakdown. Distance contributes 10% to prediction accuracy, ranking third after market and source genre popularity, and on par with target genre popularity.

Market dominates with 60% importance. Different streaming markets (US, Japan, Brazil, etc.) have fundamentally different music consumption patterns, genre preferences, and collaboration success dynamics. The model relies heavily on market information to calibrate its predictions. This finding highlights that what succeeds in one market may not translate directly to another.

The key finding for our research question is that genre distance contributes 10% to model performance, making it one of the most important predictive features after market effects and source genre popularity. Its contribution is equal to target genre popularity, suggesting that the relationship between two collaborating genres is nearly as informative as the popularity of the genres themselves.

Collaboration frequency (weight) has modest impact (6%). Whether genres collaborate frequently or rarely contributes some information, but much less than genre relationships. This indicates that the quality of the relationship (distance/compatibility) matters more than the quantity of past collaborations.

Temporal trends are minimal (3%). Success patterns are relatively stable across 2017-2019, with year contributing little to predictions. This suggests the collaboration landscape didn’t shift dramatically during this period, and the patterns we observe are relatively stable over our study window.

7 Ablation Study and Distance Effects

This section presents our main findings regarding the research question: "Does genre distance provide statistically significant predictive value in collaboration success prediction?" We test this through an ablation study, then analyze how distance contributes to predictions, the direction of its effect, and whether these findings hold across different contexts.

7.1 Ablation Study

To test directly whether genre distance provides statistically significant predictive value beyond popularity and market information, and whether hyperbolic embeddings add value beyond raw co-occurrence data, we conducted a three-model ablation study using Gradient Boosting with identical hyperparameters.

Table 8 presents the results:

Model	Features	CV R^2	Test R^2
A (Baseline)	popularity, weight, market, year	0.658 (± 0.002)	0.665
B (ENAO Similarity)	+ ENAO co-occurrence weights	0.659 (± 0.003)	0.665
C (Hyperbolic Distance)	+ Hyperbolic distance	0.677 (± 0.002)	0.684

Table 8: Ablation study results comparing three feature sets. Model B tests raw ENAO co-occurrence weights, while Model C tests hyperbolic distances derived from those weights.

The baseline model without any genre relationship feature achieves test $R^2 = 0.665$. Adding raw ENAO co-occurrence weights (Model B) provides no improvement, maintaining test $R^2 = 0.665$. In contrast, adding hyperbolic distance (Model C) improves performance to $R^2 = 0.684$, representing a 2.9% relative improvement over baseline.

The failure of ENAO similarity to improve predictions reflects ENAO graph sparsity: 88.3% of collaboration pairs have no direct edge (similarity = 0.0), meaning the feature is essentially constant for the vast majority of observations and provides the model with little useful information. Genre pairs as different as Pop $\times$ Rock and Pop $\times$ Death Metal both receive a similarity of 0.0, making them indistinguishable to the model.

Hyperbolic distance, in contrast, provides continuous variation across all 43,661 collaboration pairs. Even genre pairs with no direct ENAO edge are assigned meaningful distances based on their position in the learned embedding space, allowing the model to distinguish between stylistically close and distant collaborations regardless of whether they share a direct connection in the dataset.

7.1.1 Statistical Significance

Paired t-tests confirm the statistical significance of these performance differences:

- B vs A (ENAO vs Baseline): $p = 0.778$ (not significant, fail to reject H_0^{AB})
- C vs A (Hyperbolic vs Baseline): $p < 0.001$ (highly significant, reject H_0^{AC})
- C vs B (Hyperbolic vs ENAO): $p < 0.001$ (highly significant, reject H_0^{BC})

These results confirm that adding hyperbolic distance significantly reduces prediction error compared to both the baseline and ENAO similarity, while ENAO similarity alone provides no significant improvement. This demonstrates that the predictive value of genre distance comes from the geometric structure learned by hyperbolic embeddings, not from the raw co-occurrence patterns used to train them.

7.2 How Distance Contributes to Predictions

Having established that genre distance significantly improves prediction accuracy, we now examine how it contributes. Specifically, for which types of collaborations distance information is most valuable.

To understand where distance adds the most value, we measured the prediction accuracy improvement gained by adding distance for each collaboration. We computed prediction error from both models and calculated the difference, which indicates whether including distance made predictions more accurate or less accurate for each type of collaboration. We divided the test set into three equal-sized bins based on distance using quantiles to ensure 33.3% of samples per bin.

Table 9 presents the average error improvement by distance level:

Distance Level	Avg Error Improvement	Sample Count
Low	0.0008	2,911
Medium	0.0050	2,911
High	0.0158	2,911

Table 9: Error reduction by distance level. Distance helps prediction accuracy most for diverse collaborations with high distance.

Key finding: Distance helps most for diverse collaborations. The error improvement increases dramatically with distance level. For low-distance pairs, adding distance reduces error by only 0.0008 on average, which is a negligible improvement. For medium-distance pairs, improvement rises to 0.0050. For high-distance pairs, improvement jumps to 0.0158, which is approximately 20 times higher than for similar pairs and around three times higher than for medium-distance pairs.

This pattern reveals an important insight: genre distance provides the most value where predictions are hardest to make. Without distance information, the model struggles to predict outcomes for genre-diverse collaborations accurately. In particular, the model lacks information about genre compatibility, audience overlap, and whether the stylistic gap between genres is bridgeable.

Why does distance help less for similar pairs? For collaborations between very similar genres with low distance, the genres likely share audiences, production styles, and market positioning. Popularity and market information alone may be sufficient to predict success: if both genres are popular in a given market, their collaboration will likely succeed. Distance adds little new information because the similarity is already partially captured by correlated popularity metrics and market patterns.

7.3 Direction of Distance Effect

The previous section showed that distance improves prediction accuracy, particularly for diverse collaborations. We now examine how distance changes predictions: does it make the model more optimistic or more conservative about different types of collaborations?

For each collaboration in the test set, we computed the change in prediction caused by adding distance:

$$\text{Prediction change} = \text{prediction}_{\text{with distance}} - \text{prediction}_{\text{without distance}}$$

Positive values indicate that adding distance increased predicted success, meaning the model became more optimistic. Negative values indicate that adding distance decreased the predicted success, meaning the model became more conservative.

We analyzed how prediction change relates to genre distance by computing the correlation between distance and prediction change across all test samples.

Finding: Distance lowers predictions for diverse pairs and raises them for similar pairs.

$$\text{Correlation}(\text{distance}, \text{prediction change}) = -0.059$$

This negative correlation indicates that as genre distance increases, the model’s predicted stream counts tend to decrease. In other words, including distance makes the model more optimistic about collaborations between similar genres and more conservative about collaborations between distant ones.

Table 10 shows the average prediction change by distance bin:

Distance Level	Avg Prediction Change	Interpretation
Low	+0.003	Minimal change
Medium	+0.026	Predictions increase
High	-0.022	Predictions decrease

Table 10: How adding distance affects predictions.

The pattern shows that adding distance raises predictions for low and medium-distance pairs (with medium showing the largest effect at +0.026) while lowering them for diverse pairs (-0.022), indicating the model learns that genre similarity correlates with higher collaboration success.

7.3.1 Reconciling with Error Reduction

At first glance, this finding may seem to contradict Section 7.2, which showed that distance helps accuracy most for diverse pairs. How can lowering predictions for diverse pairs improve their accuracy?

The answer lies in understanding what the baseline model gets wrong. Without distance information, it has no way to distinguish between similar and diverse collaborations. It treats a Pop × Dance Pop collaboration identically to a Pop × Death Metal collaboration if they have similar popularity and market characteristics. As a result, the model systematically overestimates the success of genre-diverse collaborations, because it lacks the information needed to account for the challenges of bridging distant audiences.

When distance is added, the model learns to calibrate its expectations:

- **For similar pairs (low distance):** The model recognizes that shared audiences and stylistic compatibility create favorable conditions, so it raises predictions slightly.
- **For diverse pairs (high distance):** The model recognizes that audience fragmentation and stylistic gaps create challenges, so it lowers predictions.

This calibration improves accuracy by correcting the baseline model’s systematic overestimation of diverse collaborations. When distance is included, the model learns to be more cautious about genre-distant pairs, and this caution is justified. Diverse collaborations are genuinely harder to predict and, on average, slightly less successful when controlling for other factors. Distance provides information that helps the model reflect this pattern.

7.3.2 Concrete Example

Consider a diverse collaboration with high distance = 7.5:

- Actual success: 6.3 log(streams)
- Prediction WITHOUT distance: 6.8, overestimated by 0.5
- Prediction WITH distance: 6.4, overestimated by only 0.1

The prediction went down, from 6.8 to 6.4, but became more accurate, with error decreasing from 0.5 to 0.1. This is how distance helps despite lowering predictions: it provides a reality check that reduces overconfidence.

7.3.3 Interpretation: Distance as a Calibration Function

Genre distance does not have a simple positive or negative effect on collaboration success. Instead, it acts as a calibration mechanism that adjusts predictions based on genre compatibility. The model learns that similar genres are associated with higher average predicted success. However, this reflects a general tendency rather than a universal rule for individual collaborations. Its effect depends on context, interacting with factors such as market and popularity.

This finding also explains why the raw correlation between distance and success is approximately zero, as discussed in Section 5. Distance does not directly determine success through a simple linear relationship. Rather, it modulates predictions in a context-dependent way, helping the model distinguish favorable conditions, such as similar genres and overlapping audiences, from challenging ones, such as diverse genres and fragmented audiences.

From a theoretical perspective, this supports the audience overlap hypothesis: similar genres likely share fan bases, making their collaborations more predictable and slightly more successful on average. Diverse genres may face the challenge of bridging distinct audiences, introducing uncertainty, meaning they are harder to predict. The model’s behavior—lowering predictions for diverse pairs and raising them for similar pairs—suggests it has learned patterns in the training data where similarity correlates with success, though establishing causality would require comparing genre pairs with matched popularity within the same market.

7.4 Context Dependency: Market-Level Patterns

Although market contributes 60% to model performance, we have not yet examined whether distance provides consistent value across different streaming markets. Different regions have distinct music cultures, genre preferences, and collaboration patterns, which raises the question of whether genre distance matters equally everywhere or whether its contribution varies by market.

We evaluated model performance separately for each of the nine markets in our dataset by filtering the test set by market and computing R^2 for each subset.

Table 11 presents per-market performance:

Market	Test R^2	Test Sample Size
US	0.411	1,153
Global	0.417	1,205
Japan	0.416	520
Australia	0.400	1042
Canada	0.402	1,261
Brazil	0.386	609
Germany	0.349	1,011
UK	0.359	1,199
France	0.299	733

Table 11: Model performance by market. France is the hardest market to predict, with $R^2 = 0.299$, while US, Global, and Japan are easiest, with $R^2 \approx 0.41$.

Finding: Prediction difficulty varies substantially by market. R^2 ranges from 0.299 for France to 0.417 for Global, indicating that collaboration success patterns are more predictable in some markets than others. The US, Global, and Japan markets achieve R^2 around 0.41, while France proves particularly difficult to predict at $R^2 = 0.299$.

7.4.1 Why Do Markets Differ?

Several factors may explain this variation. France’s low predictability ($R^2 = 0.299$) may reflect unique cultural factors such as French-language content regulations, quotas for French-language music on radio [20], a distinct music industry structure, or genre preferences that do not align with global patterns. France’s music market has historically been more isolated, which may create collaboration dynamics that our genre-level features cannot fully capture.

The higher predictability of the US, Global, and Japan markets ($R \approx 0.41$) may reflect larger markets with more data, more stable genre ecosystems, or clearer relationships between genre characteristics and streaming success. These markets may also have more internationally-oriented music industries where genre-based predictions work better.

These market-level prediction difficulties likely reflect factors not captured by our genre-level features, such as language preferences, local artist popularity, and cultural factors specific to each market. While our ranking-based popularity metric is computed separately for each market using local Top 200 charts, it aggregates all artists from a genre without distinguishing between domestic and international acts, which may be particularly important in markets like France where local content preferences are strong [20].

The overall model R^2 of 0.684 is much higher than any individual market R^2 because the model uses market identity as a feature, allowing it to learn different prediction patterns for different regions. This improves overall accuracy but also reinforces a broader point: collaboration success is fundamentally shaped by geographic context, and what works in one market does not necessarily translate to another.

7.4.2 Distance Contribution Across Markets

While sample sizes prevent reliable market-specific feature importance estimation, distance’s 10% contribution to the overall model, which already accounts for market differences, suggests it provides value universally. If distance only mattered in a small subset of regions, its overall importance would likely be lower. The consistent contribution suggests that genre relationships remain relevant across multiple geographic contexts. This finding has practical implications. Collaboration success is highly market-dependent: the same Pop $\times$ Rock collaboration might achieve very different streaming numbers in France versus the US due to cultural preferences, language, and industry structure. However, the compatibility signal provided by genre distance- whether genres are stylistically close or distant- remains relevant across all markets. While A&R teams and recommendation systems must account for market-specific patterns in absolute success levels, genre distance offers a universal indicator of collaboration compatibility that transcends geographic boundaries.

7.5 Context Dependency: Genre Family Analysis

Beyond market-level variation, we examined whether distance provides differential value across genre families. We grouped genres into five families: Pop, Rap, Hip Hop, Rock, and Latin. We then analyzed both within-family versus cross-family collaborations and how much distance improves predictions for each family.

7.5.1 Within-Family vs Cross-Family Collaborations

We categorized each test collaboration as either:

- **Within-family:** Both genres belong to the same family, such as Pop $\times$ Dance Pop.
- **Cross-family:** Genres belong to different families, such as Pop $\times$ Rock.

Table 12 presents the comparison:

Type	Count	% of Test	Avg Distance	Avg Success	Model R^2
Within-family	605	7%	2.16	6.68	0.778
Cross-family	8,128	93%	4.35	6.59	0.677

Table 12: Within-family collaborations are easier to predict than cross-family collaborations. They have lower average distance, 2.16 versus 4.35, and higher model R^2 , 0.778 versus 0.677.

Cross-family collaborations are significantly harder to predict. Within-family collaborations achieve $R^2 = 0.778$, compared to 0.677 for cross-family collaborations. This difference is statistically significant ($t = -6.716$, $p < 0.001$) with mean absolute errors of 0.264 versus 0.338 respectively.

This pattern reflects the different prediction contexts these collaboration types represent. Within-family collaborations, such as Pop with Dance Pop, occur within a cohesive musical and market context where both genres share similar production aesthetics, target similar audiences, and operate within the same market segment. Success depends primarily on popularity and market trends, which our model captures well. Cross-family collaborations, such as Pop with Rock, bridge distinct musical traditions, different fan demographics, and potentially different market segments. This introduces both opportunities like reaching new audiences or risks like audience rejection or stylistic mismatch. While

our genre-level features capture some of this complexity through distance, the additional uncertainty from crossing family boundaries means that substantial unpredictability remains.

7.5.2 Distance Effect by Genre Family

We measured how much adding distance improves predictions for collaborations involving each genre family. For each family, we trained two models, with and without distance, on collaborations where at least one genre belonged to that family. We then compared test R^2 .

Table 13 shows the results:

Family	Test Samples	R^2 Without Distance	R^2 With Distance	Improvement
Rock	390	0.516	0.640	+0.124 (+24.0%)
Hip Hop	1914	0.447	0.506	+0.060 (+13.2%)
Rap	2,177	0.574	0.629	+0.055 (+9.6%)
Pop	2,800	0.567	0.610	+0.043 (+7.6%)
Latin	569	0.533	0.572	+0.040 (+7.5%)

Table 13: Distance improvement by genre family. Rock shows the largest absolute improvement, +0.124, while Latin shows the smallest, +0.040.

Rock is most distance-sensitive. Adding distance improves Rock collaboration predictions by 0.124, from $R^2 = 0.516$ to $R^2 = 0.640$, a 24.0% relative improvement. This is the largest gain among all families. Pop and Latin show smaller improvements, at +7.6% and +7.5% respectively.

7.5.3 Why Is Rock Most Distance-Sensitive?

The Rock family’s high sensitivity to distance likely stems from its exceptional stylistic diversity. Rock encompasses a broad range of subgenres, from soft rock and pop rock to hard rock, alternative rock, and metal. This breadth creates substantial variation in distance even within the family: Soft Rock $\times$ Classic Rock exhibits low distance, while Alternative Rock $\times$ Death Metal shows very high distance. Distance information helps the model distinguish between these vastly different collaboration types in ways that popularity metrics alone cannot capture.

This stylistic diversity corresponds to audience fragmentation. Unlike Pop, which targets mass audiences through broadly accessible sound and production, or Latin music, which maintains cohesion through shared language and cultural identity, Rock audiences tend to be subgenre-specific. Alternative rock fans may have limited overlap with classic rock fans, and metal audiences represent a distinct demographic entirely. Distance provides crucial information about whether a collaboration can successfully bridge these audience segments or will struggle to find a cohesive fan base.

In contrast, Pop and Latin collaborations appear less sensitive to distance for different reasons. Pop’s commercial orientation means success depends heavily on star power, marketing investment, and playlist placement. Two popular Pop artists collaborating will likely succeed regardless of their specific subgenre positions, making distance a secondary consideration. Latin music collaborations, meanwhile, are shaped more by linguistic and cultural factors, whether artists share Spanish or Portuguese language audiences and appeal to specific geographic markets, than by pure stylistic distance within the Latin genre taxonomy. The shared cultural context provides a foundation for success that operates independently of the genre distance metric.

7.5.4 Interpretation

Distance does not provide uniform value across all genre contexts. Its importance is modulated by the characteristics of the genre families involved. For Rock, with its wide stylistic range and fragmented audiences, distance is particularly valuable, improving predictions by nearly 24%. For Latin, with its cultural cohesion, distance matters less, improving predictions by 7-8%.

This finding suggests that genre distance captures different aspects of compatibility depending on musical context. Distance’s meaning varies by genre ecosystem: it may reflect stylistic diversity in genres with broad taxonomies like Rock [14], audience demographics and network effects in collaborative genres like Hip Hop and Rap [7], or cultural and linguistic boundaries in regionally-rooted genres like Latin [20]. The predictive value adapts to what distance represents within each genre family’s context. For Rock collaborations, genre distance is a particularly important factor, while for Latin and Pop, other considerations such as language, cultural identity, and artist popularity are likely to matter more.

8 Discussion

This study yields three key findings that form the basis for interpreting genre distance’s role in predicting collaboration success. First, the relationship between features and success is strongly non-linear, as tree-based models substantially outperform linear regression, indicating that genre distance interacts with market dynamics and popularity features in a context-dependent way. Second, genre distance is a meaningful predictor of success, contributing 10% to model performance and ranking just below market effects and individual genre popularity measures, showing that it captures relevant information beyond standard popularity signals. Third, the model demonstrates strong overall generalization performance, although a substantial share of variance remains unexplained, likely reflecting factors such as promotion, timing, and song-specific effects not captured in the data.

Taken together, these findings indicate that genre distance plays a statistically significant but structurally complex role in predicting collaboration success. To examine this relationship in more detail, this study examined whether genre distance, computed using hyperbolic embeddings learned from the Every Noise at Once genre dataset, provides statistically significant predictive value for collaboration success measured as average Spotify streams. The ablation study confirms that it does: adding genre distance to the Gradient Boosting model improves test R^2 by 0.019 and reduces mean squared error significantly ($p < 0.001$). However, this answer requires qualification. The effect is modest in absolute terms, the relationship is non-linear and context-dependent, and the raw correlation between distance and success is essentially zero. Understanding why these findings appear contradictory is the central interpretive challenge of this study.

8.1 The Core Finding: A Statistically Significant but Non-Linear Effect

The most striking result is that genre distance improves predictions despite having a near-zero linear correlation with streaming success (Pearson $r = -0.03$ globally). At first glance, this appears contradictory. However, the exploratory analysis reveals that the relationship is non-linear rather than absent. When collaborations are grouped into distance bins, a consistent inverted-U pattern emerges. Medium-distance collaborations outperform both low- and high-distance ones across most markets and years. A linear correlation coefficient cannot capture this kind of structure, which explains why

it registers near zero even when a meaningful relationship exists.

This finding is consistent with Ordanini et al. [4], who show that the effect of genre differences on chart success varies depending on the structural properties of the genres involved. Our results extend this to a streaming context and make the non-linearity explicit through both the binned analysis and the behavior of tree-based models. Linear Regression, which assumes a straight-line relationship, achieves only $R^2 = 0.149$, while Gradient Boosting achieves $R^2 = 0.684$. This gap strongly suggests that the underlying relationships in the data are non-linear and complex, requiring a more flexible modeling approach to capture them. Genre distance does not simply add or subtract a fixed amount from predicted success. Its effect depends on the market, the genres involved, and their respective popularity levels.

The results further show that distance is most valuable precisely where predictions are hardest to make. Adding distance reduces prediction error approximately 20 times more for high-distance collaborations than for low-distance collaborations. This pattern is consistent with the audience overlap hypothesis: when two genres are similar, their fan bases largely overlap, and popularity and market information alone are enough to make reasonable predictions. When genres are more distant, the model needs additional information to assess whether that gap can be bridged. Genre distance provides exactly that signal.

The comparison against raw ENAO co-occurrence weights further strengthens this conclusion. While ENAO similarity, the direct training signal for the embeddings, provides no improvement over the baseline ($p = 0.778$), hyperbolic distance significantly enhances predictions ($p < 0.001$). This demonstrates that the geometric transformation is not merely reproducing the input data but extracting higher-order structural relationships that the raw weights cannot capture. The limited predictive value of ENAO similarity can be attributed to the sparsity of the ENAO graph: with 88.3% of collaboration pairs having no direct edge, it functions as a weak signal rather than a continuous measure of genre relationships. Hyperbolic embeddings mitigate this limitation by encoding hierarchical structure across the genre network, enabling meaningful distance computation between any genre pair regardless of direct connectivity.

8.2 Interpreting the Effect Direction

The direction analysis shows genre distance acts as a calibration mechanism rather than a direct predictor of success. When distance is added to the model, it adjusts its stream predictions based on genre compatibility, predicting slightly higher stream counts for similar genres and slightly lower stream counts for distant ones. The model learns that genre similarity tends to be associated with higher success on average, while greater distance is associated with higher uncertainty, leading the model to lower predictions.

This finding aligns with intuition. Genre distance tells the model something that popularity alone cannot - whether the audiences of two collaborating genres actually overlap. Two genres that are close in hyperbolic space likely share similar fan bases, creating favorable conditions for a successful collaboration. Collaborations between distant genres may involve separate fan communities, which introduces both opportunity and risk that popularity metrics cannot distinguish between. This is why distance contributes 10% to model performance (comparable to each individual popularity metric) de-

spite having near-zero linear correlation with success. It captures a dimension of genre compatibility that no other feature in the model provides.

8.3 Market and Genre Context

The high feature importance of market (60%) shows that collaboration success is strongly shaped by geographic context. The same collaboration may perform very differently across markets, and the model relies heavily on market identity to calibrate its predictions. France stands out as a consistent anomaly. It is the hardest market to predict ($R^2 = 0.299$) and the only market showing a positive global correlation between distance and success. This may reflect the distinctiveness of the French music market. Page and Dalla Riva [20] identify France as one of the European markets most strongly characterized by local content preferences and domestic artist dominance on streaming charts, driven in part by longstanding language-based content regulations. These locally oriented dynamics may create collaboration patterns that genre-level features alone cannot fully capture.

At the genre family level, Rock benefits most from distance information (+24.0% improvement in R^2) while Latin benefits least (+7.5%). This implies that distance means different things in different genre contexts. For Rock, where subgenres cluster tightly together, even small differences in distance carry meaningful information about whether a collaboration fits within the genre or crosses into unfamiliar territory. For Latin, shared language and cultural identity likely drive collaboration success more than stylistic distance does. This is consistent with the broader finding from Ordanini et al. [4] that the effect of genre distance depends on the structural properties of the genres involved.

8.4 Limitations

Several limitations qualify the conclusions of this study. First, the dataset consists exclusively of collaborations that appeared in Spotify’s Top 200 charts, meaning all observations represent at least some degree of commercial success. Collaborations that failed or never reached the charts are entirely absent from the dataset. This means the study only observes successful collaborations, leaving unsuccessful ones invisible. The model is not learning what separates successful collaborations from unsuccessful ones, but rather what distinguishes more successful collaborations from less successful ones within an already filtered pool. As a result, the findings can only speak to relative success among already successful collaborations, not to success in absolute terms.

Second, the findings are observational rather than causal. Artists self-select their collaborators, and the genre distances observed in the data reflect human decisions rather than random assignment. It is possible that artists already intuitively optimize for genre distance, meaning the dataset is not a neutral sample of all possible collaborations. This study can show that genre distance is associated with collaboration success, but it cannot establish whether distance itself drives that success or whether both are shaped by factors not captured in the data.

Third, the study is limited to Spotify streams across three years (2017–2019) in nine markets. Music consumption patterns differ substantially across platforms, and the findings may not generalize to YouTube, Apple Music, or other streaming services. Additionally, the dataset predates major industry shifts such as the rise of TikTok as a music discovery platform and changes in streaming behavior brought by the COVID-19 pandemic, which limits how applicable the findings are to the current music

landscape.

8.5 Future Work

Several directions could extend this study meaningfully. While our ranking metric is computed per market, it could be improved by developing market-relative genre positioning that accounts for how genres compare to each other within a specific market rather than using absolute ranking scores. Additionally, distinguishing between domestic and international artist popularity within each market could better capture local music consumption patterns.

A further validation of the hyperbolic embeddings could be achieved through formal clustering metrics such as the silhouette score [34]. While the intra-family distance analysis in Section 4.1.4 shows that genres within the same family cluster significantly closer than random pairs, this comparison is informal. Computing silhouette scores across genre families would provide a standardised, quantitative measure of how well-separated and cohesive the learned genre clusters are, offering stronger evidence that the hyperbolic geometry captures meaningful structure beyond what distance comparisons alone can confirm.

Beyond this, causal inference methods could help establish whether genre distance genuinely causes collaboration success or merely predicts it. While this study shows that distance improves prediction accuracy, it cannot rule out that both distance and success are shaped by unmeasured factors such as promotional investment or artist-level characteristics. Audience overlap analysis would also be a valuable extension. By directly examining whether collaborating genres share listener bases future work could empirically test the audience overlap hypothesis that this study relies on as a theoretical explanation. Multi-platform validation using data from YouTube or Apple Music would test whether the findings generalize beyond Spotify. Finally, incorporating song-level features such as audio characteristics, lyrics, and production style alongside genre-level distance could address the 31.6% of variance currently unexplained by the model.

9 Conclusion

This study shows that genre distance, computed in hyperbolic space using the Lorentz model, provides statistically significant predictive value for collaboration success on Spotify. The effect is modest, non-linear, and strongest for collaborations that cross larger stylistic distances.

Critically, the three-model ablation study demonstrates that this predictive value comes from the geometric structure learned by the hyperbolic embeddings, rather than from the raw co-occurrence patterns used to train them. While direct ENAO co-occurrence weights provide no significant improvement over the baseline, hyperbolic distance yields a highly significant improvement ($p < 0.001$), confirming that the embedding process captures higher-order structural relationships in the genre taxonomy that simple pairwise similarity cannot provide.

The key takeaway is that genre relationships carry predictive information that popularity metrics alone cannot provide, and that this information only becomes accessible when genres are represented as continuous geometric structures rather than flat categorical labels or sparse similarity scores. For artists, producers, and streaming platforms, this suggests that the relationship between genres in a collaboration is a measurable factor in its outcome, beyond a purely creative choice.

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